

LYDIA PEDERSEN

AI PIPELINE ARCHITECT & DOCUMENTATION ENGINEER

949-466-2819 · lydiacaire.x@gmail.com · lyd1acla1r3.github.io

PROFESSIONAL SUMMARY

Documentation engineer and AI Pipeline Architect with 20 years of experience across physics research, hardware validation, and AI-powered content systems. I am driven by the challenge of translating ambiguous human intention into highly controlled, quasi-deterministic programmatic workflows. I build agentic pipelines that replace manual judgments with reliable, automated systems. Unlike standard implementations, my pipelines rely on extensive A/B testing, structured result auditing, cross-file consistency checks, and strict reproducibility. Every result is evidence-backed and mathematically calibrated to prevent regressions, applying the same engineering discipline to AI systems that hardware teams apply to certification.

SELECTED ENGINEERING PROJECTS

Transformer Architecture & AI Foundations Ebook Series

Personal Portfolio

- Authored five interconnected technical series (53 chapters, 5 ebooks) building the full inference pipeline of a Decoder-only Transformer from first principles: token encoding (BPE), vector embeddings and similarity search, positional encoding (sinusoidal and RoPE), the complete forward and backward pass (Q/K/V attention, residual stream, layer normalization, MLP), and Mixture of Experts sparse routing. Every series derives the underlying tensor algebra by hand.

AI Tooling Ecosystem & Reference Architecture Guides

Personal Portfolio

- Authored a 6-part series mapping the AI tooling landscape into functional zones, categorizing 11 tool types and establishing decision rules for assembling production systems.
- Designed and published 5 complete reference architectures with step-by-step setup guides and annotated configurations: Enterprise Agentic Pipeline (LangChain, LangGraph, Weaviate, Langfuse), Solo Developer Stack (Claude Code, Ollama, ChromaDB), RAG-First Knowledge System (LlamaIndex, Pinecone, Braintrust), Evaluation-Driven Development Loop (vLLM, Unsloth, LiteLLM), and Private/Airgapped Deployment (Qdrant, SGLang, Arize Phoenix).

PROFESSIONAL EXPERIENCE

Senior Technical Writer

Jun 2023 – Jul 2026

Gravitee · Remote

AGENTIC PIPELINE DEVELOPMENT

- **AI Doc Writer Agent:** Multi-stage content generation pipeline that converted PR diffs and Jira tickets into publication-ready documentation. Featured fidelity scoring, Best-of-N evaluation with adaptive scaling, a comprehensive guard cascade to prevent hallucination and editorial fabrication, extensive post-processing, and multi-stage recovery. Output quality was measured through containment-based deduplication metrics and cross-file consistency checks.
- **Analytics Deflection Engine:** Abstracted a monolithic codebase into a modular, product-agnostic multi-step pipeline that transformed raw AI assistant analytics into ranked, privacy-safe documentation improvement reports. Featured an extensive parameter sweep for self-correction, multi-signal priority scoring (impact, pain, breadth), a comprehensive text cleaning cascade, k-anonymity privacy model, and automated Jira task generation. Surfaced what users actually struggled with so documentation teams could close knowledge gaps systematically.
- **Link Governance Engine:** Multi-category link classifier with a safety-gated autofix system, page-version index validation, and external link health checking. Automated broken link detection, classification, and remediation across versioned documentation sites with deterministic, auditable fix decisions.
- **Semantic Redirect Engine:** Multi-stage redirect pipeline (Extract, Embed, Match, Emit) using embedding-based semantic similarity with a penalty stack, token intelligence system, and multi-tier fallback cascade. Generated intelligent cross-version page mappings that preserved user navigation across documentation releases.

AI DEVELOPMENT METHODOLOGY

- Built all pipelines through AI-assisted development using agentic and prompt engineering patterns. Optimized model selection for token budgets and rate limits. Maintained fresh context through checkpoints, structured handoffs, and living documents that preserved continuity across conversations. All work was version-controlled and followed reproducible, non-destructive workflows.

DOCUMENTATION ENGINEERING & STRATEGY

- Automated headless browser-based screenshot and UI capture, eliminating manual image maintenance. Integrated PR diff analysis for change-aware content generation synchronized with product releases.
- Elevated documentation to a first-class engineering deliverable. Coauthored team processes, templates, strategies, and roadmaps. Redesigned documentation site information architecture through goal-centric user journey analysis.

Technical Writer

Entando · Remote

Oct 2021 – Mar 2023

Owned developer documentation for a Kubernetes-based application composition platform. Verified all content in cloud and Multipass VM environments with hands-on testing of MFE/MS generation, product UIs, and CLI tooling.

- Maintained a VuePress static documentation site with Markdown/Vue styling. Created and owned the company glossary and style guide, standardizing terminology across all customer-facing materials.
- Authored a high-impact white paper used to secure new investors. Performed content and language review of marketing, developer advocacy, and partner communications. Coached non-writers in documentation best practices.

Hardware Validation Engineer

Amazon Lab126 · Sunnyvale, CA

Apr 2017 – Sep 2021

Led high-speed interface compliance testing (USB, SATA, HDMI) from proof-of-concept through production certification for consumer devices. Owned acceptance criteria, test plans, data collection, and design feedback. This is the same systematic validation methodology I now apply to documentation pipeline output.

- Developed test automation scripts and custom tooling for data collection, processing, and analysis across evolving product lines. Built repeatable, instrumentation-aware test methodologies for comprehensive coverage.
- Managed vendor relationships for test equipment procurement and support. Maintained industry engagement to track standards revisions and emerging methodologies.

Senior Hardware Engineer

Oracle · Santa Clara, CA

Jul 2014 – Mar 2017

TX/RX high-speed interface characterization: jitter analysis, signal quality measurements, and PLL tuning. Developed test scripts to automate data collection, DUT parameter control, and result formatting. Applied jitter tolerance methodologies to signal integrity analysis and design feedback. Benchtop optimization including calibration, S-parameters, and post-processing.

Senior Hardware Engineer

LSI Logic (now Broadcom) · Milpitas, CA

Apr 2008 – Jun 2014

Compliance testing across PCIe, SATA, SAS, Fibre Channel, and IEEE 802.3 standards at speeds up to 32 Gbps. Early adopter of PAM4 benchtop solutions. Contributed data collection and methodology experimentation to the IEEE 802.3bj standards committee. Developed test scripts and equipment drivers to automate measurements, stress-test functionality, and perform statistical analysis.

EDUCATION

B.S. Physics, University of California, Irvine

Graduated with Honors (3.7 GPA) · UC Regents Scholar · National Physics Honors Society · Campuswide Honors Program

2002 – 2006

TECHNICAL SKILLS

AI & ML	Agentic pipeline architecture, LLM orchestration (Claude, Gemini), prompt engineering (agentic and structured patterns), AI-assisted development, transformer architecture (embeddings, positional encoding, RoPE, Q/K/V attention, MLP, backpropagation), Mixture of Experts (sparse routing), RAG, semantic search, vector databases, fine-tuning (LoRA), model evaluation, Large Language Models (LLMs), Model Context Protocol (MCP)
LANGUAGES & TOOLS	Python, YAML, JSON, Markdown, Bash/Zsh, Git/GitHub, VS Code, Jira, pytest
DOCUMENTATION	Docs-as-code, static site generators (GitBook, VuePress), API documentation, information architecture, content strategy, style guides, versioned documentation, technical ebook authoring
DATA & ANALYSIS	Statistical modeling, parameter sweep optimization, regression analysis, benchmark design, threshold calibration, data pipeline design, privacy modeling (k-anonymity)
INFRASTRUCTURE	Docker, Kubernetes, CI/CD workflows, cloud environments, Linux CLI, headless browser automation, model serving (vLLM, Ollama), AI gateway routing (LiteLLM)